# Fair turns without clashes

## Optimization · Level 1 · Lesson 4



# In this lesson you will

- Draw clashes as dots and lines.
- Colour the dots so nothing clashes, with as few colours as you can.
- Share turns fairly.

# Key ideas

- A line means: not at the same time
- Each colour is a time
- Three tests all joined need three times
- Take turns: nobody always gets the worst

# Activity: Colour the clashes

1. Draw (5 min): Draw the eight tests as dots.
2. Join (4 min): Join the tests a pupil takes together.
3. Colour (7 min): Use as few colours as you can.
4. Share (4 min): Compare with a partner.

**Use coloured pencils, or write the slot numbers 1, 2, 3 in the dots. No device needed.**

# Let’s discuss

1. Why do Omar’s three tests need three times?
2. Why does starting with the hardest dot help?
3. Which turn at school would you share more fairly?

# Exit ticket

- A line between two dots means ___. The fewest times for our test week is ___.